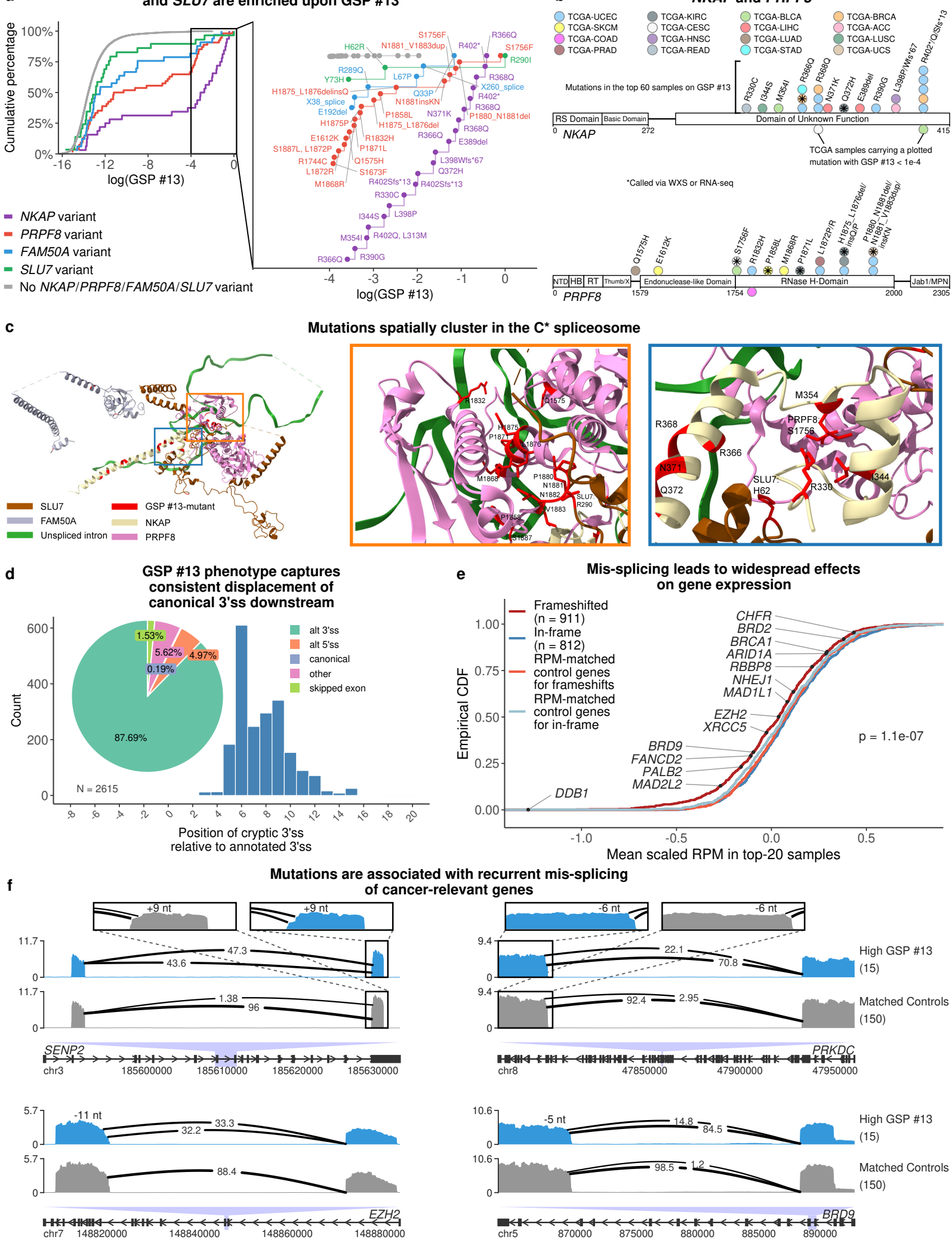

a
Somatic mutations in NKAP, PRPF8, FAM50A and SLU7 are enriched upon GSP #13
Cumulative percentage
log(GSP #13)
NKAP variant
PRPF8 variant
FAM50A variant
SLU7 variant
No NKAP/PRPF8/FAM50A/SLU7 variant
b
Mutations cluster in specific regions of NKAP and PRPF8
TCGA-UCEC
TCGA-SKCM
TCGA-COAD
TCGA-PRAD
TCGA-KIRC
TCGA-CESC
TCGA-HNSC
TCGA-READ
TCGA-BLCA
TCGA-LIHC
TCGA-LUAD
TCGA-STAD
TCGA-BRCA
TCGA-ACC
TCGA-LUSC
TCGA-UCS
Mutations in the top 60 samples on GSP #13
RS Domain
Basic Domain
Domain of Unknown Function
NKAP
TCGA samples carrying a plotted mutation with GSP #13 < 1e-4
*Called via WXS or RNA-seq
NTD
HB
RT
Thumb/X
Endonuclease-like Domain
RNase H-Domain
Jab1/MPN
PRPF8
c
Mutations spatially cluster in the C* spliceosome
SLU7
FAM50A
Unspliced intron
GSP #13-mutant
NKAP
PRPF8
d
GSP #13 phenotype captures consistent displacement of canonical 3'ss downstream
alt 3'ss
alt 5'ss
canonical
other
skipped exon
1.53%
5.62%
4.97%
0.19%
87.69%
N = 2615
Count
Position of cryptic 3'ss relative to annotated 3'ss
e
Mis-splicing leads to widespread effects on gene expression
Frameshifted (n = 911)
In-frame (n = 812)
RPM-matched control genes for frameshifts
RPM-matched control genes for in-frame
Empirical CDF
Mean scaled RPM in top-20 samples
p = 1.1e-07
CHFR
BRD2
BRCA1
ARID1A
RBBP8
NHEJ1
MAD1L1
EZH2
XRCC5
BRD9
FANCD2
PALB2
MAD2L2
DDB1
f
Mutations are associated with recurrent mis-splicing of cancer-relevant genes
+9 nt
-6 nt
High GSP #13 (15)
Matched Controls (150)
SENP2
PRKDC
-11 nt
-5 nt
EZH2
BRD9